Examiner only

10. *(a)* (i) Tin and carbon are both more stable in oxidation state +4 whilst lead is more stable as +2. Explain the difference in the stabilities of these oxidation states. [1]

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(ii) Lead(IV) oxide can act as an oxidising agent when it reacts with concentrated hydrochloric acid. Write an equation for this reaction. [1]

....

(b) Tin(II) ions can be used to reduce iron(III) ions in **acidic** solution.

$$2Fe^{3+}(aq) \quad + \quad Sn^{2+}(aq) \longrightarrow 2Fe^{2+}(aq) \quad + \quad Sn^{4+}(aq)$$

$[Fe^{3+}]$ / mol dm^{-3}	$[Sn^{2+}]$ / mol dm^{-3}	pH	Initial rate / mol dm^{-3} s^{-1}
0.200	0.100	0	1.4×10^{-2}
0.300	0.200	0	2.8×10^{-2}
0.400	0.100	1	1.4×10^{-3}
0.400	0.200	1	2.8×10^{-3}

(i) Find the rate equation for this reaction. [3]

(ii) Calculate the value of the rate constant, *k*, for this reaction. [2]

k = ...

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(iii) A student suggests that the rate determining step for this reaction is:

$$Fe^{3+} + Sn^{2+} \longrightarrow Fe^{2+} + Sn^{3+}$$

State, giving a reason, whether this is a possible rate determining step for this reaction. [2]

....

....

....

(c) The rate of this reaction can be followed using sampling and quenching.

(i) Explain what is meant by *sampling and quenching*. [1]

....

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(ii) A 5.00 cm^3 sample of the solution was analysed by titration against acidified potassium manganate(VII). The sample required 27.20 cm^3 of manganate(VII) solution of concentration 0.00205 mol dm^{-3} for complete reaction.

Calculate the concentration of Fe^{2+} in the solution. [2]

$$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$$

$$Fe^{2+} \longrightarrow Fe^{3+} + e^-$$

Concentration of Fe^{2+} = ... mol dm^{-3}

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